Code No: 157EZ **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
REMOTE SENSING AND GIS
(Common to EEE, CSE, IT)

Time: 3 Hours **Max.Marks:75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What are the advantages of sensors carried on board satellites?	[2]
b)	What are the four types of images? Give their applications.	[3]
c)	What is the relation between GIS and SDSS?	[2]
d)	What are the GIS layers used in land use mapping?	[3]
e)	What do you mean by normalization? List its advantages.	[2]
f)	Explain metadata and its uses.	[3]
g)	Write a note on errors in data input and its effects.	[2]
h)	What do you mean by data accuracy and how do you ensure the same?	[3]
i)	What is the importance of decision making in GIS?	[2]
j)	How do you demarcate hazard zones using GIS?	[3]

PART – B

(50 Marks)

2. Present a detailed classification of remote sensing systems and explain their merits and demerits. [10]

OR

3.a) Explain the elements in evaluating images.
b) Write a detailed note on visual interpretation techniques. [5+5]

4.a) Explain the importance of maps in GIS.
b) What is a digital terrain model? Explain the role of digital terrain model in selection of dam site. [5+5]

OR

5.a) What is map standardization? Explain how it is carried out.
b) Write notes on Scale; Coordinate systems; Geo-referencing. [5+5]

6.a) Bring out a detailed comparison between object and field-based models.
b) Write a detailed note on Topologically Integrated Geographic Encoding and Referencing. [5+5]

OR

7.a) Explain different types of raster data structures.
b) Explain topology, topological and non-topological data structures. [5+5]

8.a) Explain different methods of data input and explain their merits and demerits.
b) Write a note on Micro and Macro components of accuracy. [5+5]

OR

9.a) Explain the salient aspects of vector data analysis.
b) Discuss the various techniques used for spatial data interpolation. [5+5]

10.a) Write a note on system requirements for implementing a GIS.
b) Discuss how decision making helps in implementation of GIS. [5+5]

OR

11.a) Explain how Remote Sensing and GIS techniques are used in identification of sites of artificial recharge in groundwater development
b) Explain how GIS is useful in identification of shortest route in Municipal Solid waste collection. [5+5]

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